

CTD60M – Probe



Manual and operating instructions



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Sea & Sun Technology GmbH / Erfurter Str.2 / D-24610 Trappenkamp / Germany
Tel.: ++49 (0) 4323 910913 / Fax: ++49 (0) 4323 910915 / www.Sea-Sun-Tech.com

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PH/ORP

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1. General description

The CTD60M memory probe is a small and handy microprocessor controlled multiparameter probe for precise online measurements as well as for self-contained operation in and shallow water. Apart of the CTD sensors it is possible to interface a number of chemical, physical and optical sensors and instruments. The housing of the probe is pressure resistant up to 2000m depth, the maximum allowable operation depth depends on the sensors line-up. Due to its handiness and low weight of less than 3 kg it is particularly suited for portable employment without the use of winches. The probe is completely made of titanium (except screws). The housing is inert against nearly all chemical compounds (except hydrofluoric acid) and absolutely corrosion free.

The small housing diameter of 60 mm allows a maximum of 5 sensors to be connected to the bottom cap. Additional sensors or instruments can be attached externally to the probes top cap via underwater connectors (maximum 2 external instruments).

The standard CTD60M allows operation in different modes:

- Time mode
- Increment mode
- Continuous mode
- Online (FSK or RS232C)

Time mode is the most important mode for long term measurements, where the operator selects start and stop time and the measuring time interval.

Increment mode is mainly used during profiling and enables the user to carry out a great number of profiles without reading out the stored data files in between. After configuration of start and stop depth and the measuring depth increment the probe will store one data set at each predestined depth.

In online mode the CTD60M runs on standard single conductor cables with constant current, the measurement readings are transmitted as FSK-signals

modulated on the DC supply current. This method of operation requires a specific probe interface which generates the constant current and converts the FSK-signals into PC-compatible RS232-signals. This mode of operation allows the releasing of multi water samplers and plancton multi nets and to recognize the status of these units (number of bottles closed or number of net changes).

For shorter distances (several hundred metres) a multicore cable can be used. The probe is then supplied with constant voltage (battery or DC power supply). The PC received data directly from the probe as RS232C-signal. A specific interface is not required by this mode of operation.

The CTD60M is equipped with a 16 channel data acquisition system with 16 bit resolution. A high long-time stability and automatic self-calibration of the 20 bit analogue digital converter guarantees stable and precise CTD measurements for many years.

2. Mechanical characteristics

All parts of the probe which are exposed to seawater are made of corrosion-proof metals or plastic. Essentially the probe consists of the following mechanical structural components:

- Pressure pipe
- Probe base
- Probe lid with battery pack
- Sensor protection cage
- Sensors

The sensors are described later in a separate chapter.

2.1. Pressure pipe

The pressure pipe is made of a solid-drawn seamless titanium tube with an external diameter of 60 mm and a wall thickness of 3,9 mm and is able to withstand more than 2000 m water depth. There are 4 holes in 90° graduation in the side of each tube-end. These holes are used for fixing both end caps of the housing to the pressure tube with 4 screws M3 * 6. Pressure pipe and end caps are sealed by two O-rings 48 * 2,5 mm each.

2.2. Probe base

The probe base is the bottom end cap, made of titanium, which is used for the fixation of all CTD sensors and cable connections for external sensors and units. The base has room for maximum 5 sensors or connectors. The

pressure sensor is always mounted in the centre. All sensors (except the pressure sensor) have identical flanges. The pressure transducer has thread 1/4" UNF 28THD as calibration connection to a pressure gauge so that the sensor can be calibrated when installed.

Figure 1 CTD60M bottom view

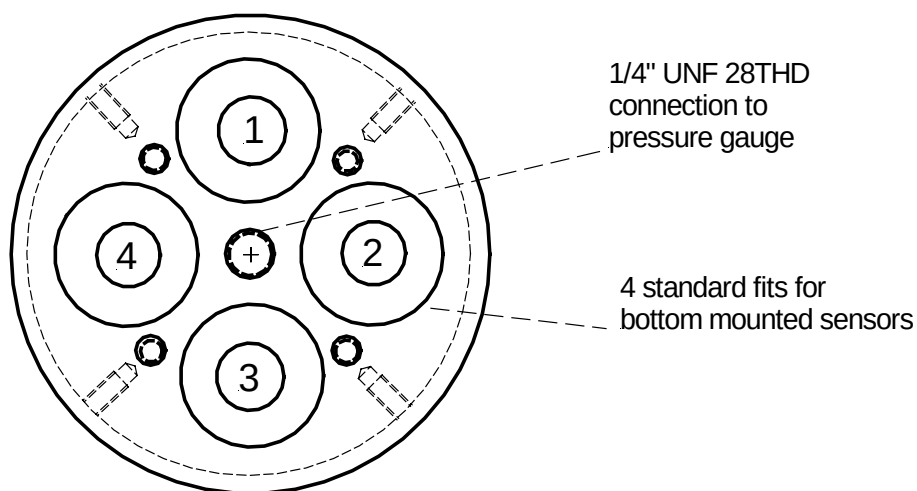
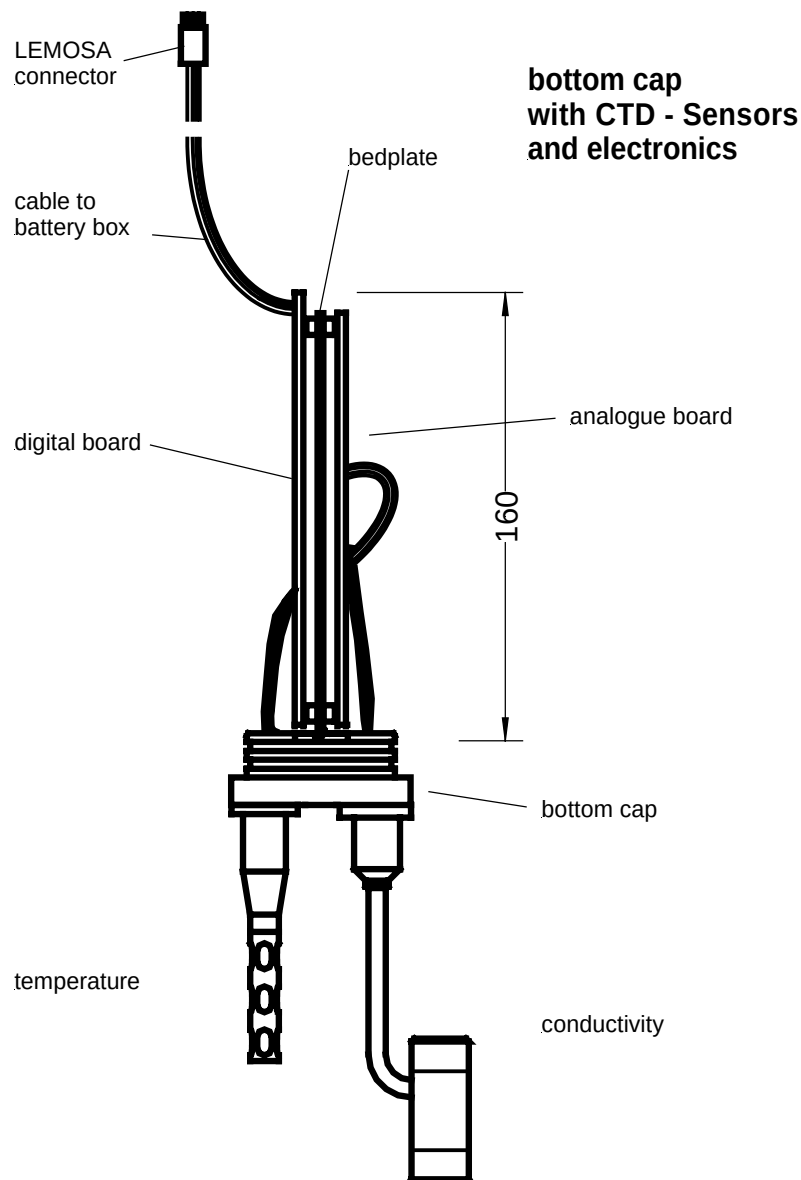


Photo shows CTD with Seapoint Turbiditymeter



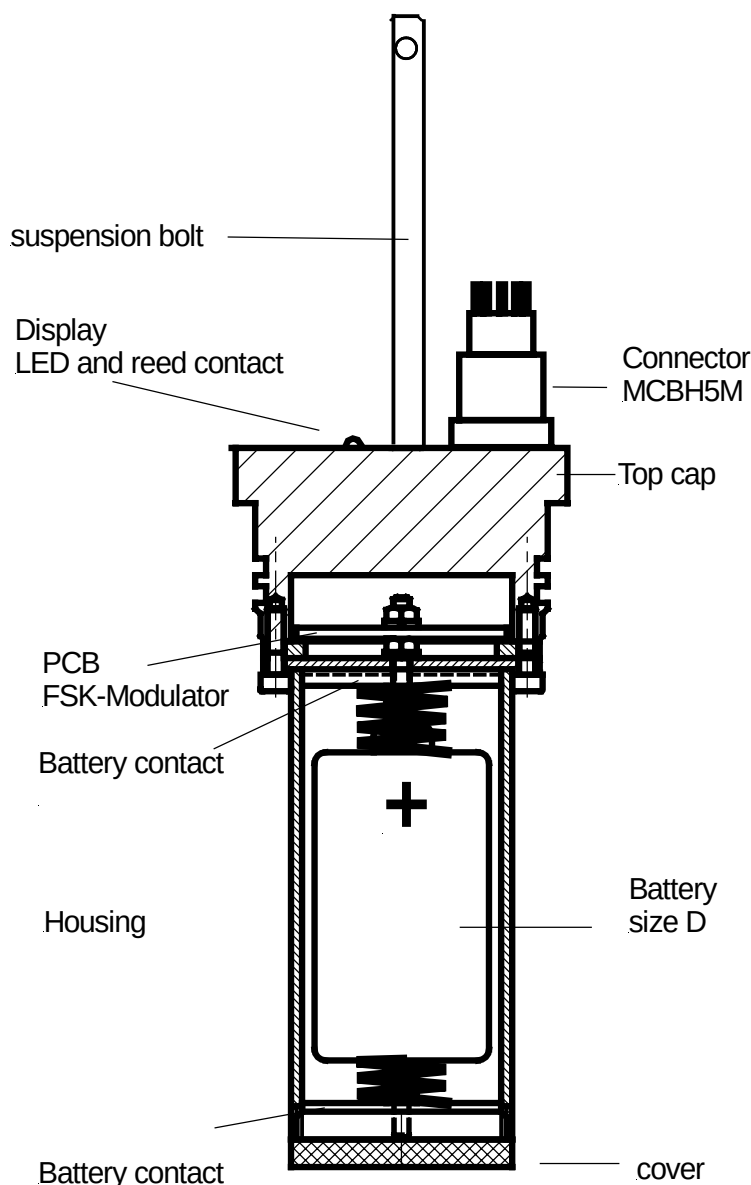


On the inside of the bottom cap a bedplate mounting support is vertically fastened. It is made of 2 mm thick aluminium sheet and bears both of the probes printed circuit boards (digital board and analogue board).

Base and pressure tube are sealed by two O-rings 48* 2,5mm and are screwed together with 4 screws M3 * 6.

2.3. Probe lid

The lid has the same dimensions as the base and is also made of titanium. Fastening and sealing are identical to that of the base. Screwed into the lid is a supporting bolt with a loop for hanging it onto a shackle. The standard version includes one underwater bulkhead connector SUBCONN MCBH5M which is used for communication (configuration and data readout) and external power supply and one operating control device. The control device consists of a red LED for the display of operation conditions and messages and a magnetic switch (reed contact) to turn on and off the instrument.



A circular board is mounted on the inside between lid and battery box. It contains the the cabel driver and FSK modulator and all the necessary wiring of connectors, control device and battery box. The connection to the probe electronics is established by a separable cable connection.

2.4. Sensor protection cage

A sensor protection cage made of 6 mm titanium rods with a diameter of 120 mm and a length of 220 mm is delivered with the standard version. The protection cage protects the sensors on the water-floor against shocks and ground contact and guarantees a fine water-flow through the sensors. The protection cage is fastened with a single screw at the lower end of the pressure pipe. As option any other size can be supplied.

2.5. Dimensions and weights

pressure tube

material:	titanium
length:	340 mm
diameter:	60 mm
wall thickness:	3,9 mm
depth capability:	2000 m
weight:	1kg

bottom cap:

material:	titanium
diameter:	60 mm
thickness:	30 mm
depth capability:	2000 m
weight (incl CTD sensors and electronics):	0,7kg

top cap:

material:	titanium
diameter:	60 mm
thickness:	30 mm
depth capability:	2000 m
weight (including battery pack):	1 kg
weight with batteries:	1,1 kg

protection cage

material:	titanium rod 6mm
diameter:	120 mm
length:	220 mm
weight:	0,3 kg

standard probe:

gross length:	650mm
total weight:	< 3 kg
bouyancy:	1 kg

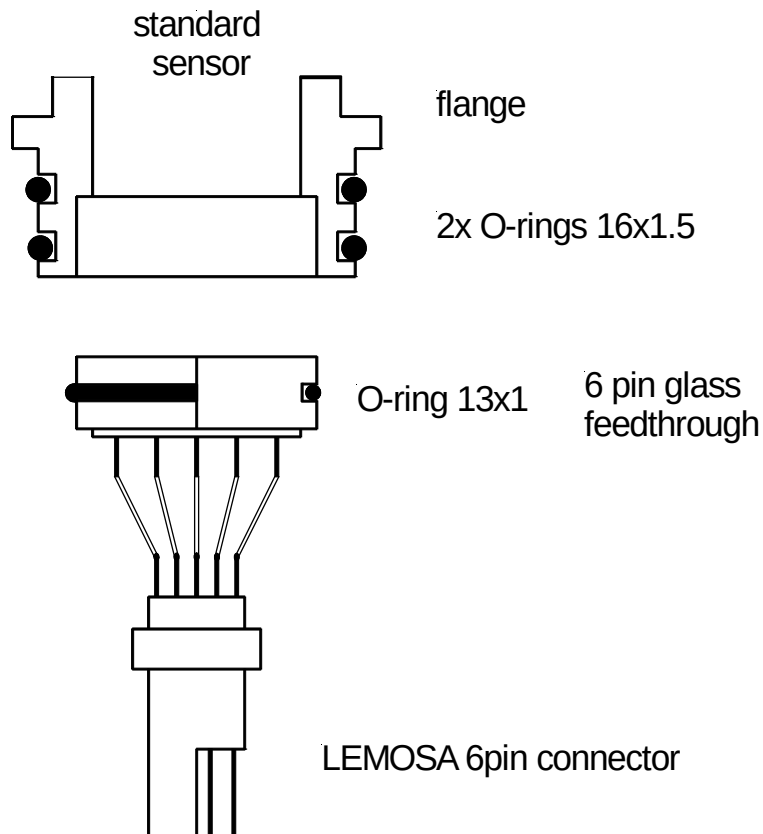
3. Sensors

The CTDM 60 has a 16 analogue channels and 8 digital inputs or outputs for the connection of different sensors. A maximum of 5 sensors fit onto the bottom. The other sensors or instruments have to be connected externally via additional underwater plugs in the sensor lid.

The following sensors can be accommodated in the sensor-base (bottom mounted sensors without cable connection)

- Pressure transducer
- Ground runner
- Temperature sensor Pt 100
- Conductivity cell
- Oxygen sensor
- Ph and redox sensor
- Seapoint turbidity meter

The standard sensors have the same flange with an integrated six-pin glass feed through equipped with a small six-pin round connector (see figure below). All of these sensors can therefore be removed from the outside and can easily be replaced without having to open the probe.



External sensors

with analogue outputs and cable connection to the top cap of the CTD90

transmissiometer

light sensors (LI-COR)

fast oxygen sensor (AMT)

H₂S sensor (AMT)

can be attached and operated as external sensors. Power for external sensors has to be supplied by the CTD90. Standard supply voltage is 12 volt.

* Please note: The TC-sensor flange is not equipped with a glass feedthrough due to the high number of wires ($T = 4 + C = 6$)

3.1. Pressure transducer

A piezo-resistive full bridge in OEM version with a diameter of 15 mm and a total height of 5,6 mm is used as pressure transducer (produced by the Swiss manufacturer KELLER). The casing and diaphragm are made of alloy C-276. The transducer is delivered with a small SMD-PCB and includes a temperature compensation of the pressure measurement. The sensor is mounted in the base of the probe; the SMD-board has contacts and is plugged onto the main board of the probe.

Manufacturer:.....KELLER, Switzerland

Type:.....PA7-20 Progress (*)

Measurement range:.....XXX Bar (*)

Burst pressure:..... 150 %FS

Overall accuracy:.....0,1%FS

Sensor diameter:.....15 mm

Sensor height.....5,6 mm

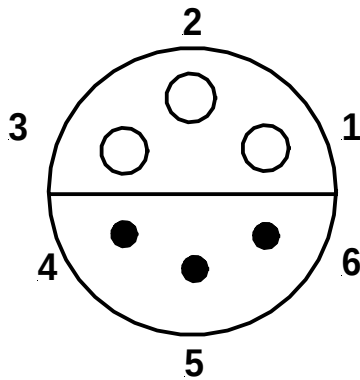
O-ring:.....13*1 mm

(*) = 5, 10, 20,50,100, 200 Bar = **Full Scale**

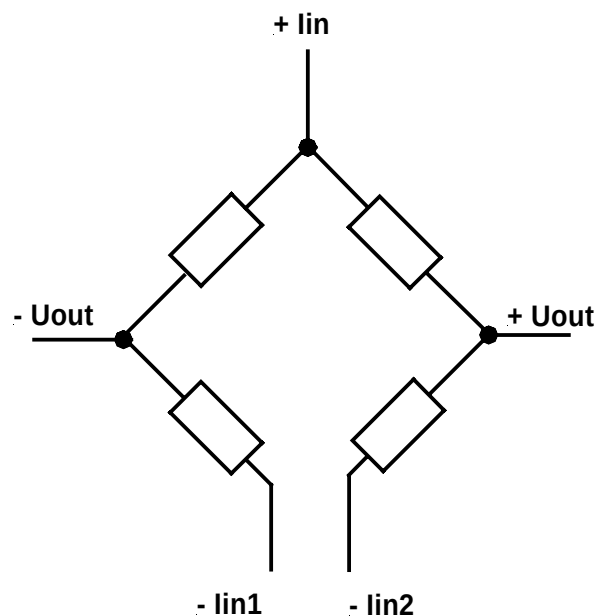


The sensor is a full wheatstone bridge with a bridge resistance of 3,5kOhm at air pressure and driven by constant current. The actual temperature of the bridge silicon chip is measured via the bridge voltage and used for compensation of thermal drifts of zero point and sensivity. This results in good temperature and long term stability.

A female thread $\frac{1}{4}$ "UNF 28THD with a depth of 10 mm in the center of the bottom cap allows the connection to a pressure gauge for calibration. Please take care that the maximum allowable thread length of the gauge connector does not exceed 10 mm.



**Lemosa connector
PSA.1S.306.ZLL
face view**



Pin-No.	Signal	cable color
1	+ I in	black
2	- I in 1	white
3	- U out	blue
4	+ U out	red
5	- I in 2	yellow
6	not connected	

3.2. Temperature-Conductivity sensor



3.2.1. Pt 100

is a platinum resistor of 100 Ohm (at 0°C) and a size of 0,9 mm diameter and 15 mm length. It is mounted into a thin titanium tube of 1,2 * 0,1 mm and approximately 35 mm length. The fine needle-shaped pipe is very sensitive against touching or bending and therefore sheltered by a perforated protecting tube of 10 mm diameter. The electrical connection is done in four wire technique in order to avoid the influence of the wire and the connector resistances.

Manufacturer:.....SST

Sensor element:.....ISOTECH P 100/1509

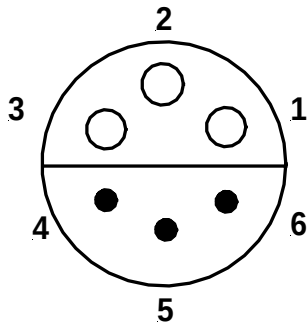
Time constant:.....150 msec at 1m/sec flow

Length:.....90 mm

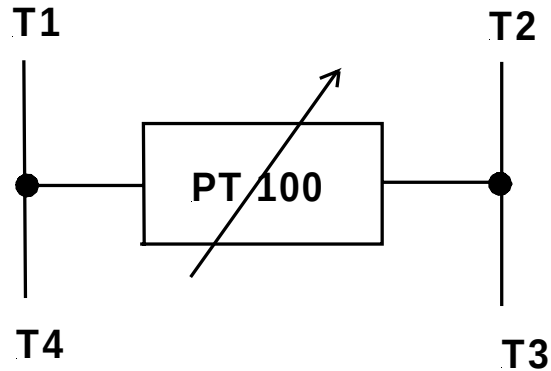
Connector:.....Lemos PSA.1S.306.ZLL

Pin No.	Signal	Colour
---------	--------	--------

1	T1	black
2	T2	white
3	T3	white
4	T4	black
5	not connected	
6	not connected	

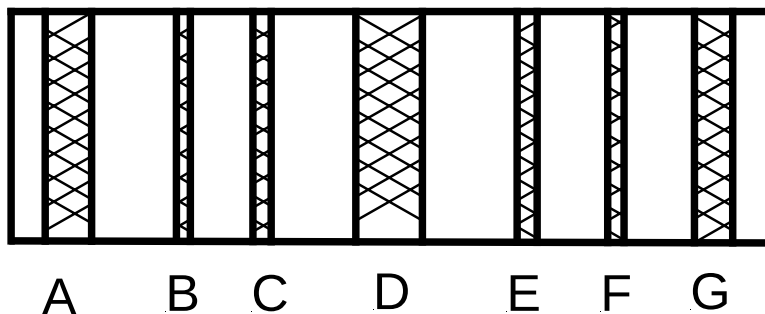


**Lemosa connector
PSA.1S.306.ZLL
face view**



3.2.2. The conductivity sensor

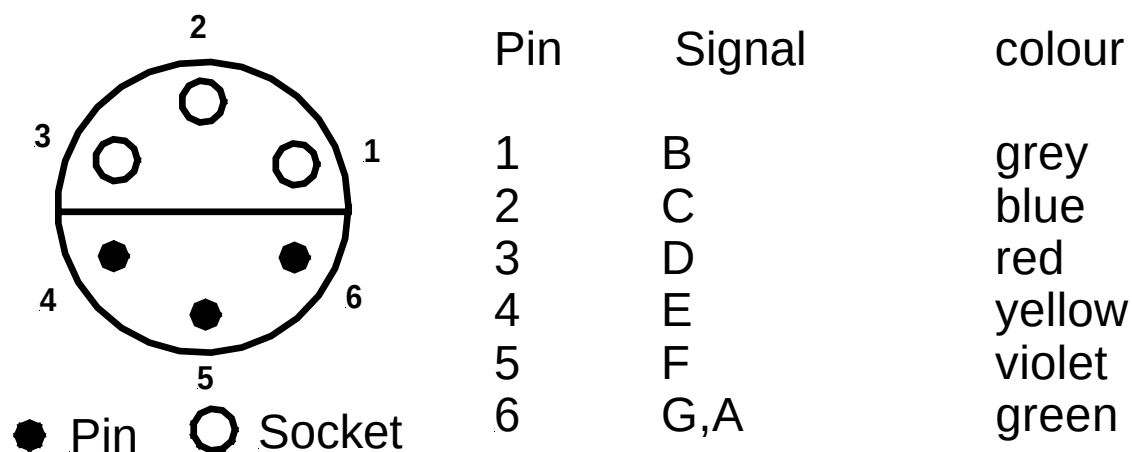
uses a cylindrical 7 electrode quartz glass cell with platinum coated rings as electrodes in the inner side of the glass cylinder. The following schematic shows the electrode arrangement and cell geometry .



The outside and middle electrodes are the current electrodes, the smaller ones are the voltage sensing electrodes. The middle electrode (D) is driven by an alternating current while the outside electrodes (A,G) are held constant on sea water potential. This symmetrical arrangement leads to an independance of the conductivity measurement from boundary conditions because the electrical field is completely kept inside the glass cylinder. The great advantage is an easy and accurate calibration procedure with the same results in any surrounding. The housing of the conductivity cell is made of a special sealing compound founded in a mould.

Manufacturer:AMT
 Type:.....7 electrodes cylindrical cell
 max. depth:..... 200 Bar
 Ranges:.....0 – 1 mS/cm, 0 - 6 mS/cm, 0 - 60 mS/cm (*)
 Length:.....110 mm
 Connector:..... Lemos PSA 1S.306.ZLL

(*) any range between 1...70mS/cm is possible



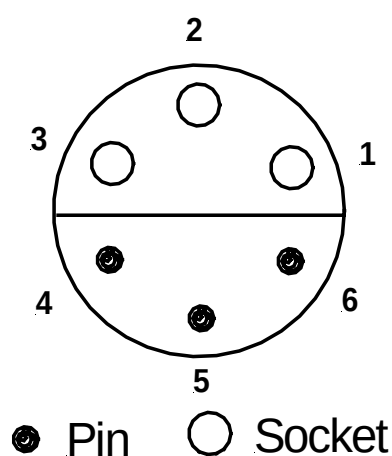
3.4. Oxygen sensor

The oxygen sensor measures the dissolved oxygen in the water using polarographic methods. The platinum cathode has a diameter of 4mm and is encased with a teflon membrane. The oxygen current consumption ranges from 0 to 12 μ A due to the big diameter of the platinum wire. The relative high current consumption requires a minimum current flow of 10 cm/sec in order to avoid oxygen depletion in front of the membrane.



Technical data:

Manufacturer	SST/Oxyguard DO522M18
Type	Clark electrode, self galvanizing
Polarisation voltage:	-0,7 VDC
Range	0 – 150 %
Oxygen current	0 – 12 nA
Temperature range	-2°C – 30°C
Response time	approx. 3s (63%), 10s (90%)
Accuracy	+ /-3%
Maximum depth	2000 m



Pin	Signal	colour
1		
2	+Vout	brown
3	- Vout	blue
4		
5		
6		

3.5. pH and Redox sensors

3.5.1 Depth range 0..160m

pH and redox combined electrodes are industrial sensors using a solid reference system (stiff polymer mass containing KCl) and an aperture diaphragm which allows direct contact between reference electrolyte and sample medium. Regeneration of the glass membrane or filling up electrolyte is not possible. When the lifetime of the sensor is over, it has to be replaced by a new one. The sensor has a thread PG 13,5 and is screwed into a flange. A coaxial socket makes the electrical contact in the flange. Sealing between sensor and flange is achieved by an O-ring, which is part of the sensor.

Technical data:	pH	Redox
Manufacturer	METTLER-TOLEDO	METTLER-TOLEDO
Model	405-DXK-S8/120	Pt 4805-DXK-S8/120
Measuring range	4-10	-2000mV.. – 2000 mV
Maximum depth	160m	160 m
Shaft diameter	12 mm	12 mm
Length with flange	167 mm	167 mm
Response time	approx. 1 sec	approx. 1 sec.

pH sensor (160 m)



3.6.2. Depth range 0..500 m

pH and redox combined electrodes based on the same principles as described in §3.6.1 but more pressure resistant.

Technical data:	pH	Redox
Manufacturer	Hamilton	Hamilton
Model	Polylite PRO 120 XP	Polylite RX 120 XP
Maximum depth	500m	500m

Other technical data same as above.

PH sensor (500m)



3.6.3. Depth range 1200m

This pH/ORP Sensor uses a pressure-balanced glass electrode with a reference to provide in-situ measurements up to 1200m depth. The sensor is equipped with a reference system using a solid gel (stiff polymer mass containing Ag^+ -free KCl) and a ceramic pore diaphragm and with a pressure stable pH-sensitive glassy electrode. The pH probe is permanently sealed and supplied with a soaker bottle attachment. The bottle contents must be 3 mKCl solution (pH 4) that prevents the reference electrode from drying out during storage.

This kind of sensor is absolutely **H₂S resistant**.

	PH	Redox
Manufacturer	AMT GmbH	AMT GmbH
Measuring range	4-10	-2000mV.. – 2000 mV
Maximum depth*	1200m	1200 m
Shaft diameter	12 mm	12 mm
Shaft material	transparent plastic	transparent plastic
Bulkhead material	Stainless steel	stainless steel
Thread	G1/4 (ISO228)	G1/4 (ISO228)
Shaft length	84mm	84mm
Length with flange	117 mm	117 mm
Response time	approx. 1 sec	approx. 1 sec.



This sensor is pressure resistant up to several thousand meters depth with a slight increase of pH/ORP values.

3.7. Seapoint turbidity sensor

The bottom mounted turbidity sensor is based on the SEAPOINT turbidity meter in the bulkhead version, which is screwed onto a standard flange. Electrical connection is achieved by a separable 6 pin round connector. For further details please refer to SEAPOINT's manual.

The Turbidity sensor measures the concentration of suspended matter. It is equipped with a pulsed infrared light transmitter and detects the scattered light from the particles suspended in water. Transmitter and detector arrangement uses 90° scattering at a wavelength of 880 nm. The output signal is proportional to the particle concentration in a very wide range. For detailed description of Seapoint turbidity meter refer to the special user manual.

Specifications:

Power: 7 – 20 VDC, 3,5 mA average
Signal: 0...5 VDC (each range)
Scatterance angle: 90° avg. (15...150°)
Light source wavelength: 880 nm
Linearity: 2%
Depth capability: 6000 m
Size: 2,5 cm diameter, 11 cm length



The turbidity sensor is available in two different versions: **standard** version has an underwater plug, is connected to the probe via a 6-wire cable and has to be fixed to the probes protection cage with a clamp. The **Bulkhead** version is plugged into a fit of the bottom cap of the probe and hence needs no underwater connection cable.

4. Replacement of sensors

When replacing a sensor the probe generally doesn't have to be opened (exception: pressure sensor). Proceed as follows:

- remove both of the M4-screws which hold the flange
- carefully remove the respective sensor whilst gently turning it out of its fitting in the base
- disconnect the plug contacts (pull lightly).

Reassemble in opposite order. To remove the pressure sensor the probe has to be opened. This is done in the following order:

- remove the protection case
- take the lid off: first of all unscrew the 4 M3-screws on the side of the tube-end and then pull the lid off whilst gently turning it without tilting it detach the base from the tube (as with the lid)
- disconnect all of the sensor plugs, unsolder the pressure sensor cable on the main board
- detach the bedplate from the base, unscrew the pressure sensor holding screw
- pull the sensor out carefully by its cable (from 100 m range upwards blow it out, if necessary, with compressed air from the front side)

Attention: When replacing the pressure sensor the progress-print must always be replaced as well because it contains the temperature compensation for the specific pressure sensor. When inserting the new pressure sensor grease the O-ring thoroughly. Reassembling is done in the opposite sequence.

5. Probe electronics

The electronics of the base version consists of three printed circuit boards

-
- Battery pack board (FSK modulator and cable driver)
- Analogue main board with plug-in modules
- Digital main board

5.1. The analogue main board

measures 150 mm * 50 mm and contains the following circuitry

- microprocessor controlled data acquisition
- clock
- 16 channel analogue multiplexer
- Temperature module
- Conductivity module
- Pressure amplifier
- Oxygen amplifier
- Redox amplifier
- PH amplifier
- Differential amplifiers for sensors with analog output

The main board has an expansion plug which contains all necessary signals for a system extension to 32 sensors. On the backside of the bed plate the digital board can be attached which incorporates the electronics for further sensors.

The heart of the analogue board is a microprocessor controlled 20 bit analogue-digital converter which generates an auto calibration cycle each time the probe is switched on. This results in an exceptionally good long-term stability. This is especially important for the stability and precision of the CTD sensors.

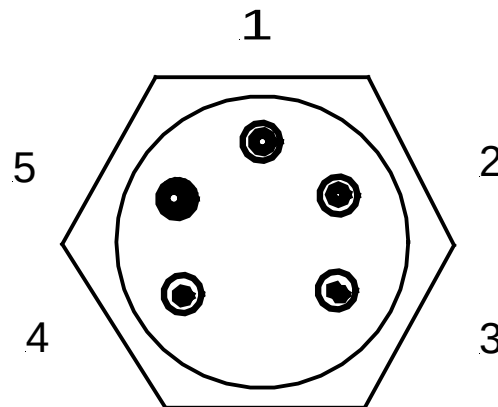
5.2. Digital board

measures 50 * 160 mm and is connected with the analogue board by a 36 pole double row connector and to the battery pack by a 10 pole round connector. It provides following functions:

- Internal voltage conversion (5 V, -5V, 3,3V)
- Power switch for external units
- Voltage supervisory circuitry
- Serial port driver/receiver
- 8 bit parallel port out
- 8 bit parallel port in
- flash eprom memory 32 Mbyte / 64 Mbyte
- Interface for magnetic switch
- Status register for all external interfaces

6. Power supply and interfaces

The CTD60M has a 5-pin underwater connector which allows the probe to be operated in different modes. The standard connector is SUBCONN MCBH5M made of titanium and neoprene.



Face view of bulkhead underwater connector
Connector pin assignment:

- Pin 1
- Pin 2 TxD, transmit data RS232C
- Pin 3 Power GND
- Pin 4 +Power input
- Pin 5 RxD, receive data RS232C

Mating cable is Subconn MCIL5F with locking sleeve MCDLSF

6.1. Internal battery

The battery box is a circular shaped housing of 40 mm diameter and approximately 90 mm height and mounted on the top cap of the memory probe. The box is designed for a single lithium battery of size D. This battery has the highest energy density of any available primary battery. All battery contacts are springs which are loaded, they assure a safe operation without power interruption even under stress and shock conditions and rough handling. The input voltage ranges from 1.5 VDC to 5 VDC.

Specifications:

Battery:	3,6 volt D cell
System	Lithium thionyle chloride Li/SOCl ₂
Nominal capacity:	16,5 Ah
Nominal discharge current	6 mA
Max. cont. discharge current	200 mA
Max. pulse current	450 mA
Size designation:	LR20 (AM1, MN1300, MONO)
Size:	33 * 62 mm
Weight:	approx. 100 g
Temperature range:	-55...85°C
Capacity loss at room temp.:	less than 1% per year

Warning

Fire, explosion and severe burn hazard. Do not recharge, disassemble, heat above 100°C, incinerate or expose contents to water!

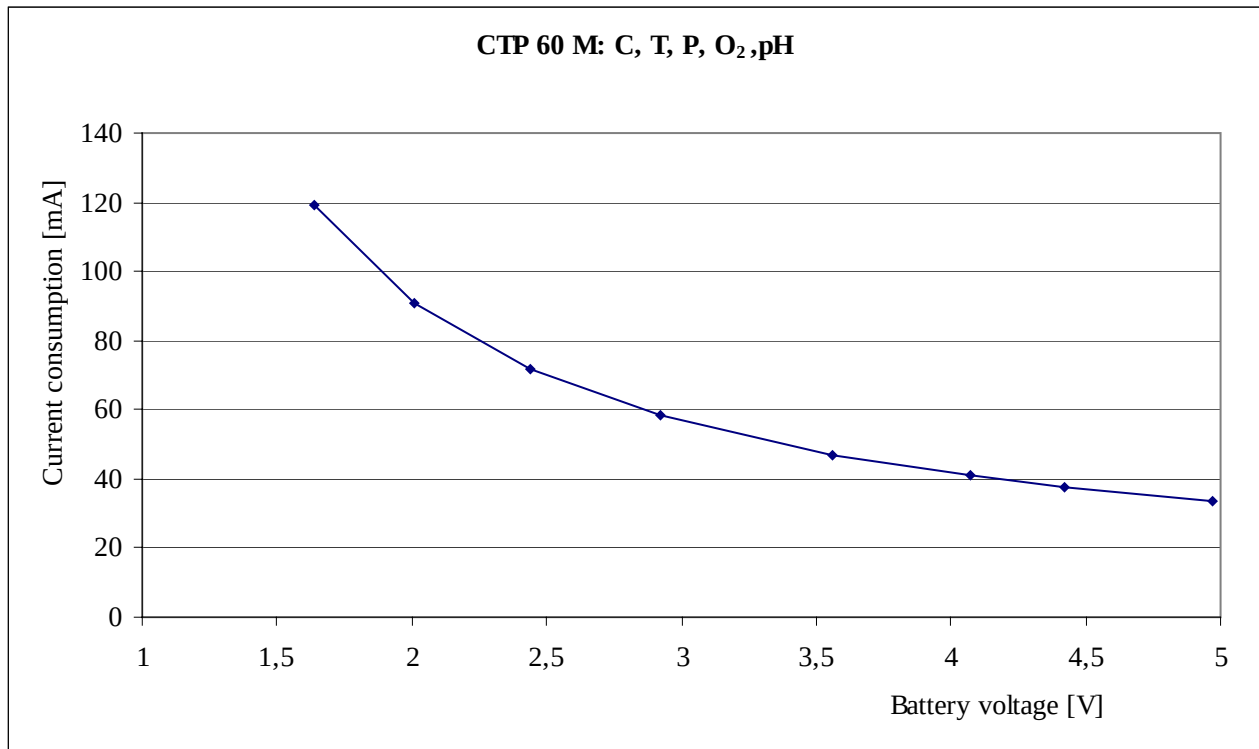
The available battery capacity depends strongly on the discharge conditions. For continuous discharge the max. capacity is 16,5 Ah at 6 mA current. the following table depicts the relation between continuous discharge current and max. available battery capacity at room temperature

0,2 mA	14,4 Ah
0,6 mA	15,0 Ah
2 mA	16,0 Ah
6 mA	16,5 Ah
20 mA	14,8 Ah
60 mA	10,0 Ah
200 mA	5,0 Ah

The capacity is also a function of ambient temperature. For currents above 6 mA the max. capacity is available at temperatures about 30 °C. It drops approximately -10% when the ambient temperature falls to -5 degree C (-20% at -20°C).

The output resistance of a lithium battery is relatively high compared with other primary batteries. Due to the static load we have a constant voltage drop of 0,2 VDC at 60 mA and 0,3 VDC at 200 mA in continuous operation. Immediately after power on we find a remarkable additional dynamic voltage drop (e.g. at 60 mA -0,3VDC and -0,8 VDC at 200 mA) which vanishes within the first ten seconds of operation. This is the reason for waiting at least 5 seconds after power on before starting data acquisition.

The following table show the probe current consumption as function of the battery voltage. When the voltage of the Lithium battery drops below 2 V, the microcontroller sets a **battery empty flag** and the power will be switched off.



The continuous operating time with a new lithium battery between 3,6 and 2 VDC will be approximately 150 hours.

6.2. External power supply

The memory probe can be powered externally via cable connection to an external battery or a DC power supply (regulated or unregulated). The external supply voltage should preferably range from 6 to 15 VDC. The internal battery need not to be removed, it is automatically switched off by the external voltage. The memory probe is then supplied by the external source via the underwater connector. Switching is executed without interruption of the internal supply voltage and data loss.

In the off-state the memory probe's current consumption is reduced to approximately 400µA drawn either from the lithium battery or from the external source.

Please note:

The maximum allowable supply voltage is limited to 15 VDC. The current consumption is approximately 40mA.

External Power Connection:

Pin 3 Power GND

Pin 4 Power In (5...15 VDC)

6.3. Online operation with multiconductor cables

The use of multicore cables is advisable for shorter distances between probe and PC and particularly in laboratory. The probe is then supplied either by battery or an external power supply (refer to 6.2.). Data transfer to the PC is via pin2 (Transmit data TxD) and pin3 (GND). A simple data line is achieved by using 3 wires, when the probe is supplied by internal battery. The RxD line is necessary for activating data transmission to the PC.

memory probe		PC / laptop	
inline cable	Signal	Signal	ser. port connector
Pin 2	TxD	RxD	Pin 2 (SUB D 9)
Pin 3	GND	GND	Pin 5 (SUB D 9)
Pin 5	RxD	TxD	Pin 3 (SUB D 9)

The maximum length of the multiconductor cable data link depends mainly on the cable resistance and capacitance and can at best be several hundred meters. An advantage is that a specific interface between probe and PC is not necessary.

6.4. Configuration cable

It is delivered with the memory probe and is intended to be used for all kind of communication between probe and PC:

- Configuration of the operation modes
- Data readout of stored files
- online transmission of data in laboratory

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The length of the configuration cable is max. 5 m, the wiring is described below:

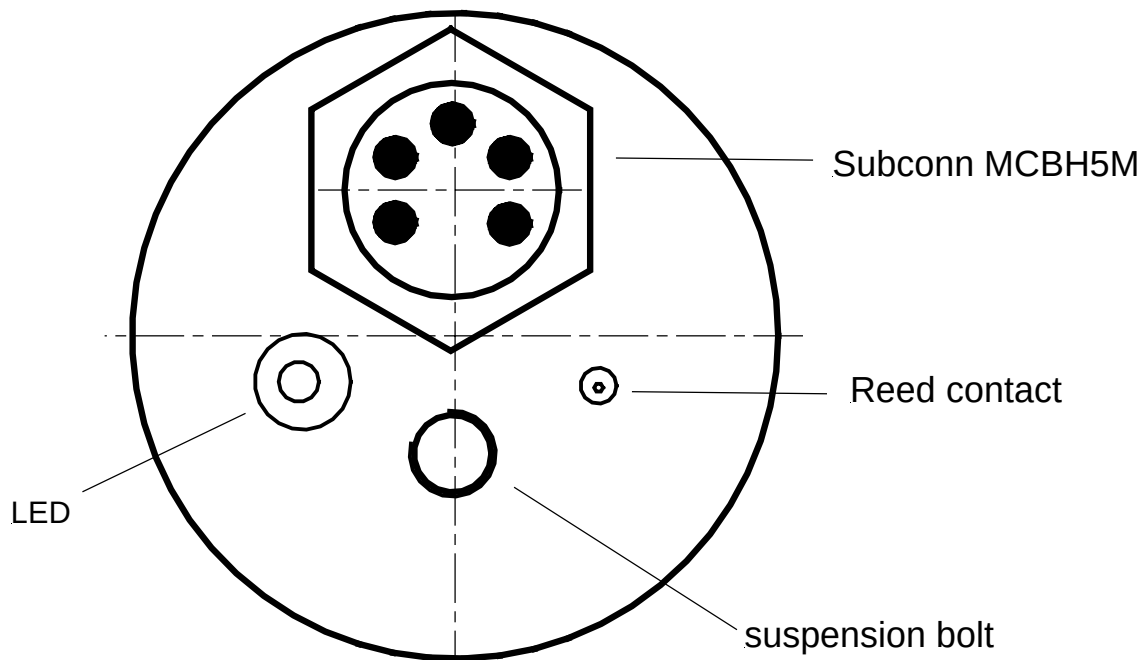
memory probe		PC serial port / Power supply	
Pin 2	TxD	RxD	Pin 2 (9 pole SUB D)
Pin 5	RxD	TxD	Pin 3 (9 pole SUB D)
Pin 3	GND	GND	Pin 5 (9 pole Sub D)
Pin 3	Power GND	Power GND	Banana plug black -
Pin 4	Power in	Power out	Banana plug red +

The configuration and data readout of the CTD60M should always be executed with the short configuration cable due to the high baudrate on the serial port (115200). The use of longer cables lead to a significant error rate during data transfer.

7. Operating the memory probe

7.1. Control elements

On the top cap of the memory probe there is a red LED and a reed contact (see drawing).



The LED is used to display operating conditions:

- red LED on.....Power on
- red LED off.....Power off
- red LED blinking.....Power on + write access to memory

Reed contact:

The activation of the reed contact is made by a magnetic rod (part of the delivery). The rod should be led vertically with the magnetic tip to the front of the mark on the end cap for not more than a second. The **ON**- condition is displayed by the red LED. A second activation turns the power off (red LED off).

7.2. Operation modes

The probe has 3 configurable data storage modes :

- time mode
- increment mode
- continuous mode

and one non configurable mode:

- RS 232 on-line mode (no data storage)

The first 3 operation modes can be configured by the supplied Windows(tm) software package "Sea & Sun Technology's **Standard Data Acquisition**". See separate manual for a complete software description. Generally the memory probe has to be configured by use of this software prior to data storage applications. Data readout and conversion to ASCII-files is done by this software, too. In addition to storage modes the probe can be used like a standard direct reading probe using RS232 data transmission. During all data storage modes the measured data is transmitted via the RS232 output to a connected PC. To save battery power the RS232 output is powered down if no valid RS232 voltage level is present at the RXD line of the probe.

Power switching

1. After being first time connected to any power source (either battery or external power supply) the probe performs a power on reset (hardware and software reset) and starts data transmitting in the non configurable mode. For this online mode the magnet has to be used to switch the probe off. Activation of the reed contact turns the power on and off (toggle switch) in all operation modes except time mode.
2. A signal on the RxD line turns the power on in any mode. The **start communication** or **restart probe** command in the SDA user menu is used to switch the probe on and **interrupts the current operating mode**.
3. Detection of turn on time (time mode) switches the power on.
4. Detection of **battery empty** flag switches the power off

7.2.1. Time mode

Time mode is configured by several parameters to best suit the data acquisition tasks of the application. The most obvious is the time interval between two consecutive wake-up periods. The second is the length of time the probe is switched on after wake-up. There is a "**Start time**", where the first interval starts and an optional "**Stop time**" after that the probe will terminate the time mode. During the **On Time** either all datasets are stored or only those at a defined time grid. The parameters for **Start Time** and **Interval** are mandatory, all others are optional for convenience.

7.2.2. Increment mode

Mostly this mode is used to obtain depth profiles with datasets stored at user-defined depth levels in order to achieve appropriate data reduction. The "**delta interval**" between two consecutive depth levels has to be entered. Optionally a **start depth** and/or a **stop depth** can be defined. After crossing one of those depth levels one complete dataset is stored and the internal processor of the probe calculates the next depth level to be crossed. Even if the same limit is crossed again later on, no additional data is stored!

More than one profile can be obtained without the necessity for data readout to a PC in between. A maximum of 250 files can be stored in this mode as long as the capacity of the internal solid state memory is not exhausted (8 Mbytes). For each profile the probe has to be switched on by use of the magnetic rod. At that moment the next file is created in the memory. If a stop depth was defined the probe will automatically switch-off at that depth and close the current data file. Otherwise the probe has to be switched off manually when it is raised to the surface (and switched on again for the next profile and data file).

72.3. Continuous mode

At this mode all acquired datasets are stored in the internal memory of the probe. Each time the probe is turned on by use of the magnetic rod a new (additional) file is created in the probe and all further datasets are stored until the probe is switched off again.

7.2.4. RS 232 on-line mode

is an operation mode without valid configuration. The memory probe enters this mode after power on under following conditions:

- After each power on reset
- The stop-time in time mode operation is reached.
- The internal data storage memory is full
- Further data storage is disabled by the PC-software

No data storage is possible, the probe awaits another configuration and acts like a direct reading probe after being switched on by the magnetic rod. The probe can operate in this mode for unlimited time until it is switched off by the magnetic rod or the battery is empty..

7.2.5 Command mode

When the PC-software is urged to "**Start communication**" with the memory probe then it switches the probe into the command mode to start configuration or readout data. During command mode no data is acquired and any data storage in progress is temporarily disabled and will be resumed after end of command mode (if wanted). If time mode is active during command mode the time-grid updates are done in background, but no data acquisition is performed! Time mode will resume after leaving the command mode and start at the next configured interval in the future.

During data readout the communication baudrate is (automatically) set to 115200 Baud to speed up the process.

Command mode is left by closing the configuration window of the PC software.

8. Service and maintenance

The best maintenance for the probe is to handle it with care. Despite the fact that the probe is sturdily designed, unnecessary strains like impacts and shocks should be avoided. Apart from that, there are only few instructions and maintenance rules which should be heeded or met to, so as to ensure a longer life-span and correct measuring results.

8.1 The underwater connector

is actually maintenance-free. However it has proved itself to be advisable to lubricate the sealing surfaces of the pins with sea water-proof grease. This reduces wear whilst plugging and unplugging. Further tips:

- Clean the plugs with warm soapy water. They do not have to be dried. Chemicals should be avoided.
- To avoid corrosion never plug or unplug whilst under water
- To conserve the cable plug never unplug by pulling on the cable. Avoid bending radiuses and above all narrow, sharp kinks.
- Plugs that are not in use should never be left blank. They should always be protected against corrosion by a dummy cap.

8.2 Pressure transducer

The pressure sensor doesn't require special attendance or maintenance. Personal experience has shown however, that the pressure sensors should never be tested by pressing any object onto the membrane. This often causes damage to the membrane, which can lead to pressure reading mistakes or to a total failure. **Pressure sensors damaged in such a way are not covered by the guaranty.**

8.3 The temperature sensor

The temperature sensor is maintenance free. Dirt and plant cover only prolong the time constant but have no effect on the precision. When cleaning the sensor take special care of the sensitive tip which should not be bent.

8.4 The conductivity cell

is principally not maintenance free. It must regularly be inspected for plant cover and electrolytic calcification. Both effects reduce the measured conductivity. It is appropriate if the probe is rinsed on deck with fresh water (better: distilled water) after each application. This reduces the formation of salt crystals on the cell surface. Calcareous deposits which originate from the electrical current flow in the cell are easily removed if the cell is immersed for a few seconds in a diluted acetic acid. If this should not be enough, repeat the procedure or use diluted hypochlorid acid. The quantity of rising CO₂-bubbles gives information on the rate of calcification. The cell is completely decalcified when the bubble formation has ceased. Afterwards the cell has to be rinsed with fresh or distilled water. Depending on the operating time this procedure has to be repeated monthly.

Cleaning is more difficult after long-time application especially during warm months, when heavy sea-pest growth densely populates the cell within a short time (2 weeks). In this case the cell has to be placed into diluted acetic or hydrochloric acid (if necessary for a longer time). This procedure may have to be repeated until the cell is completely cleaned. Then the cell is rinsed with fresh or distilled water. Particular care has to be taken, that the metal components on the electrode surfaces are not scratched, nor must they come into contact with other metals. Otherwise the life time of the cell and the long-time stability of the conductivity measurements will be impaired. After the electrodes have been treated with acetic or hydrochloric acid a short-term increased conductivity reading may occur, this should normalise within an hour.

8.5. Oxygen sensor

The oxygen sensor requires some attention from time to time. All the necessary maintenance like exchange of electrolyte and membrane is described in an OxyGuard leaflet in the appendix of this manual.

The red O-ring has two different positions:

1. in the **front position** (shown in the picture below) the O-ring prevents leakage of the electrolyte through the thread during storage. **This position should not be used for measurements but only for storage.**



2. in the **backward position** it allows the electrolyte to build a high impedance electrolytic connection between medium (sea water) and electrolyte room behind the membrane. This connection is **necessary for proper measurements. Please take care that during measurements the O-ring takes always the backward position**



The Oxyguard DO sensor is supplied by us with a sensor protection cap made of plastic . To achieve a tight fit to the sensor head the cap is equipped with an o-ring 21*1 mm and a 2mm hole in the center of the bottom (see photo). The cap should be used as protection for the membrane and sensor head as well as useful tool for oxygen field calibration.



If the membrane tension is dropping during operation or time the sensors output signal is changing too. The zero point of the oxygen sensor remains fix during its lifetime but the sensivity (slope) can vary. The user can execute a field calibration after each membrane exchange or when he doesn't trust the measured values anymore.

Field calibration

The SDA software offers the possibility to perform a field calibration and to change the reading automatically. Let the SDA program run with the probe connected to the PC. The field calibration procedure is very simple:

- Keep the membrane of the DO sensor dry
- Put the red o-ring in the backward position
- plug the protection cap onto the sensor head with a proper fitting o-ring
- Fill a small plastic cup with water and immerse the sensor head up to the flange (small white plastic cup is part of the delivery)
- after a short time the enclosed air in the cap is water vapour saturated and the the oxygen reading should have 100% partial pressure.
- If the oxygen reading is stable click menu point **Calibrate** and **02 Field Calib**
- When **02 Field Calib** is selected, the current oxygen reading is automatically stored. The default value **100%** is accepted when clicking on the button **Calculate slope now**.
- The SDA programm calculates the new oxygen Field calibration coefficient (originally 1) and the reading is now 100%.

The field calibration method works in any basin or tank and the result is independent of the salinity. When putting the complete probe into a basin you have to estimate the immersion depth of the oxygen sensor (measured from the membrane to water surface). Every 10 cm immersion depth lead to an increase of the oxygen reading of 1%. So e.g. if the procedure is executed with the DO sensor 30cm below the water surface, the default value in the button field **Enter desired value** has to be changed to 103%.

8.6. pH-sensor (ORP sensor)

Do never touch the sensitive tip. Protect the pH-sensor with the delivered soaker bottle containing the storage solution and avoid any dry out of the sensitive tip.



Avoid any air inside the bottle, fill completely with 3 M KCl. Make sure, that only 3 M KCl with pH 4 buffer is used for storage. It is not allowed to use other wetting caps in order to avoid any air pressing into the diaphragm leading to sensor malfunctions or damage. Damage because of using other wetting caps or storage without any wetting cap is not covered by guarantee. The pH sensor has to be rinsed carefully with fresh water after finishing the measurements.

The pH sensor is a replacement part and has to be changed, if the sensor has reached the lifetime. The sensor has a stainless steel thread G1/4A (titanium on request) which is screwed into a flange. The electrical contact is made by a socket in the flange. Sealing between sensor and flange is achieved by an O-ring which is part of the sensor. After the sensor's life span has ended, the sensor has to be replaced. When unscrewing the sensor, avoid that moisture (e.g. water drops) may reach the contacts (dry beforehand). A single drop of saltwater is enough to cause long-lasting incorrect measurements – this is due to the high output impedance of 100 – 400 MΩ. So only replace sensors under clean and dry conditions please.

8.7. Seapoint turbidity meter

The turbidity sensor has to be cleaned from time to time. Especially the optical sensitive flat surfaces have always to be kept clean. Avoid the use of chemical solvents.

8.8. Exchange of the battery

If the probe is prepared for use, the battery has to be put into the battery box. This has to be done in the following order:

1. Remove the four M3 screws at the upper end of the tube.
2. Pull the top cap carefully out of the pressure tube
3. Disconnect the cable connection between battery box and electronic.
3. Unscrew the top cover of the battery box
4. Take out the old battery, insert the new one.

Please observe the graphic instructions on the battery box. Inserting the battery with wrong polarity may lead to destruction of electronic components. Insert the battery with the positive pole ahead.

5. Close the top cover of the battery box and plug connectors
6. Close the top cap of the probe.

Please be very careful when closing the top cap. Use always grease for the O-ring and insert the top cap very slowly in order not to shear the O-rings at the four mounting holes of the tube.

Important hint

The cap of the battery box is the negative pole of the battery and power GND of the memory probe. The probe is supplied with power as soon as the cap get in contact with the battery box (when cable to electronic is not disconnected) or during plugging the connector again. Multiple contacts during plugging or bouncing during the the screw in procedure have to be avoided and sometimes result in a faulty reset condition and malfunction of the memory probe. **After insert of a new battery the user should check therefore the proper function of the probe:**

- LED should be ON
- Data transmission should be in online mode with all sensors transmitted
- The user can stop and restart the probe with the magnetic switch

If any of the 3 conditions is not fulfilled the user has to pull out the battery and insert it once more. **If the user is not able to insert the battery without bouncing problems it is advisable to execute the battery exchange procedure with leaving the external power ON .**

9. Online data format

The probe data can be fed into any serial ports of the PC. The standard settings of the probe are:

Baud rate	1200 or 2400
Character length	8
Number of stop bits	1
Parity	odd
Protocol	non, asynchronous
Signals	GND, TxD, RxD

Data is transmitted in binary format. Each sensor is transmitted with 3 bytes (16 bit measuring values, 5 bit address and 3 bit status). The transmission format is presented in the following chart:

Sensor	LSB							MSB
1. Byte	H	D0	D1	D2	D3	D4	D5	D6
2. Byte	H	D7	D8	D9	D10	D11	D12	D13
3. Byte	L	D14	D15	A0	A1	A2	A3	A4

DO – D15 16 bit binary data (decimal value 0 – 65535)

AO – A4 5 bit binary address (decimal sensor address 0-31)

H, H, L 3 status bits.(H = logic HIGH level, L = logic LOW level)

Transmission starts with the first byte (LSB first) and ends with the third byte (MSB last). Every sensor in the probe has a specifically assigned binary address. Address and corresponding measuring value are transmitted in a 3 byte block. The status bits are useful for the PC data acquisition programme to compose the 3 bytes in the correct sequence.

A complete data set begins with the lowest address and ends with the highest address. All addresses between 0 and 31 may occur. The transmitted physical addresses are identified by the data acquisition program and compared to those registered in the configuration file. As example the addresses for the memory probe CTD60M are listed below:

Address 1	battery voltage
Address 2	pressure
Address 3	temperature
Address 4	conductivity
Address 5	dissolved oxygen
Address 6	pH
Address 10	Turbidity

10. Calculation of the physical data

Data transmission and data storage are performed solely in binary data. The calculation of the physical values (C,T P) from the raw data and their display is carried out by the PC-data acquisition program. The calculation of physical values for standard sensors is made by a polynomial of n^{th} order:

Measurement value: $= \sum A_i * n^i$

n = raw data counts $0 < n < 65536$

A_i = calibration coefficients

$i = 0 \dots 5$

Normally $i_{\text{max}} = 1$ or $i_{\text{max}} = 2$. The calibration constants are determined by the manufacturer by comparison measurements against a normal or subnormal and subsequent regression calculations.

Further calculations such as salinity, density and sound velocity are carried out with the current UNESCO-formulas.

11. Spare parts

11.1. Sensors

- pH sensor 160m METTLER-TOLEDO HA405-DXK-S8/120
- Redox sensor 160m METTLER-TOLEDO Pt4805-DXK-S8/120
- PH sensor 500m HAMILTON PolyLite PRO 120 XP
- ORP sensor 500m HAMILTON PolyLite RX 120 XP
- PH and ORP 1200m AMT
- Pressure sensor KELLER PA7-XXX Progress 0,1 - 2 Volt
(XXX Full scale range in bar)

11.2 O rings

- Base and lid 48 * 2,5 mm
- Sensors (flange) 16 * 1,5 mm
- pressure sensor 13 * 1 mm
- PH/Redox sensor 12 * 1,5 mm
- Underwater connector 12,42 * 1,78 mm
- Oxygen cap 21 * 1mm

11.3 Plugs and cables

- Dummy cap SUBCONN MCDC5F
- Locking sleeve SUBCONN MCDLS-F
- Inline connector SUBCONN MCIL5F

12 EC declaration of conformity

Sea & Sun Technology does not provide the CTP60 probe with the CE sign but takes the right (in accordance with the national EMC law §4.2 and 4.3) to fix the EC sign in the operating instruction manual. A copy of the declaration of conformity is added to this manual in the following appendix.



Konformitätserklärung Declaration of Conformity



gemäss DIN 45014 und ISO/IEC Leitfaden 22
in accordance with DIN 45014 and ISO/IEC guide 22

Die Firma:
The company:

Sea&Sun Technology
Erfurter Strasse 2
24610 Trappenkamp

Erklärt, dass das Produkt:
declares that the product:

CTD60M, CTD90M
CTD60M, CTD90M

Verwendungszweck:
Product description:

CTD Sonde für Seewasser
multiparameter profiler for seawater

den folgenden Richtlinien der Europäischen Union entspricht
complies with the following EC guidelines

73/23 EWG Niederspannungsrichtlinie
73/23 EEC Low voltage directive

89/336 EWG EMV-Richtlinie
89/336 EEC EMC directive

Das Gerät erfüllt im einzelnen die folgenden Normen
The instrument meets the requirements of the following norms

DIN EN 55022
DIN EN 55024

Trappenkamp, 08.05. 2002

Kurt Holtsch, Techn. Leiter
Kurt Holtsch, engineering Leader